

UniDyn--Demo-08.nb

John A. Marohn
jam99@cornell.edu
Cornell University

Abstract: This demonstration notebook performs a quantum optics calculation.

Set the path to the package

Check the Mathematica version number .

```
In[68]:= $VersionNumber
```

```
Out[68]= 12.3
```

Tell *Mathematica* the path to the directory containing the packages and the /studies directory.

EDIT THE FOLLOWING PATH STRINGS:

```
In[69]:= $UniDynPath =
```

```
"/Users/jam99/Dropbox/MarohnGroup__Software_Library/UniDyn/unidyn";
```

YOU SHOULD NOT NEED TO EDIT ANYTHING FROM HERE ONWARDS.

Load and test the packages

Load Unidyn

Append the package path to the system path. Before trying to load the package, ask *Mathematica* to find it. This is a test that we directed *Mathematica* to the correct directory. The output of this command should be the full system path to the UniDyn.m file.

```
In[70]:= $Path = AppendTo[$Path, $UniDynPath];  
FindFile["UniDyn`"]
```

```
Out[71]= /Users/jam99/Dropbox/MarohnGroup__Software_Library/UniDyn/unidyn/UniDyn.m
```

Now that we are confident that the path is set correctly, load the **Unidyn** package. Setting the global \$VerboseLoad variable to True will print out the help strings for key commands in the package.

```
In[72]:= $VerboseLoad = True;  
Needs["UniDyn`"]
```

Run the Unidyn unit tests

Carry out the unit tests for the **Unidyn** functions.

```
In[74]:= SetDirectory[$UniDynPath];
```

```
In[75]:= fn = FileNames["*-tests.m"];
```

```
test$report = TestReport /@ fn;
```

```
TableForm[Table[test$report[[k]], {k, 1, Length[test$report]}]]
```

Out[77]//TableForm=

TestReportObject [	 	Title: Test Report: Comm-tests.m Success rate: 100% Tests run: 23	]
TestReportObject [	 	Title: Test Report: Evolver1-tests.m Success rate: 100% Tests run: 14	]
TestReportObject [	 	Title: Test Report: Evolver2-tests.m Success rate: 100% Tests run: 13	]
TestReportObject [	 	Title: Test Report: Evolve-tests.m Success rate: 100% Tests run: 10	]
TestReportObject [	 	Title: Test Report: Inv-tests.m Success rate: 100% Tests run: 24	]
TestReportObject [	 	Title: Test Report: Mult-tests.m Success rate: 100% Tests run: 18	]
TestReportObject [	 	Title: Test Report: OpQ-tests.m Success rate: 100% Tests run: 21	]
TestReportObject [	 	Title: Test Report: Osc-tests.m Success rate: 100% Tests run: 22	]
TestReportObject [	 	Title: Test Report: SpinBoson-tests.m Success rate: 100% Tests run: 13	]
TestReportObject [	 	Title: Test Report: Spins-tests.m Success rate: 100% Tests run: 14	]

Evolver settings

Choose which **Evolver** to use.

```
In[78]:= Evolver = Evolver2;
```

Two two-level systems coupled to a common cavity

Clear all the symbols; start afresh.

```
In[79]:= Clear[Ix, Iy, Iz, Ip, Im, Sx, Sy, Sz, Sp, Sm, aR, aL];
Clear[ω, ω$I, ω$S, g$I, g$S, Δ, n];
Clear[H$0, H$1];
```

... Clear: Symbol Im is Protected.

Create pseudo-spin operators for an I atom and an S atom.

Create raising and lowering operators for the single-mode cavity.

```
In[82]:= TwoSpinBoson$CreateOperators[Ix, Iy, Iz, I$P, I$m, Sx, Sy, Sz, S$p, S$m, aR, aL];
```

... SpinSingle\$CreateOperators: Spin operators already exist.

... SpinSingle\$CreateOperators: Adding spin commutations relations.

... SpinSingle\$CreateOperators: Angular momentum L = 1/2. Adding operator simplification rules.

... SpinSingle\$CreateOperators: Spin operators already exist.

... SpinSingle\$CreateOperators: Adding spin commutations relations.

... SpinSingle\$CreateOperators: Angular momentum L = 1/2. Adding operator simplification rules.

... OscSingle\$CreateOperators: Oscillator operators already exist.

... OscSingle\$CreateOperators: Adding oscillator commutations relations.

... TwoSpinBoson\$CreateOperators: Creating operators.

... TwoSpinBoson\$CreateOperators: Adding Ip, Im and Sp, Sm commutations relations.

... TwoSpinBoson\$CreateOperators: Adding Ip, Im and Sp, Sm simplification rules.

... TwoSpinBoson\$CreateOperators: Adding aL and aR normal ordering rule.

Scalar parameters appearing the Hamiltonians .

```
In[83]:= CreateScalar[
  ω, (* cavity frequency *)
  ω$I, (* atom I frequency *)
  ω$S, (* atom S frequency *)
  g$I, (* unitless coupling constant characterizing
    the interaction of atom I with the cavity mode *)
  g$S, (* unitless coupling constant characterizing the
    interaction of atom I with the cavity mode *)
  Δ (* resonance offset *)
]
```

Hamiltonian for the bare atoms I and S and the cavity.

```
In[84]:= H$0 = ω$I Iz + ω$S Sz + ω Mult[aR, aL];
```

Coupling Hamiltonian describing the exchange of photons between atom I and the cavity and atom S and the cavity.

```
In[85]:= H$1 = g$I ω (Mult[I$p, aL] + Mult[I$m, aR]) + g$S ω (Mult[S$p, aL] + Mult[S$m, aR]);
```

Version 0

Show that Evolver2 (also, I have tested, Evolver1) cannot evolve the initial density operator below under the full Hamiltonian.

```
In[86]:= Evolver[H$0 + H$1, t, Mult[1/2 + Iz, 1/2 + Sz]]
```

... Evolver2: Unrecognized evolution

```
Out[86]:= { { 1/4 + Iz/2 + Sz/2 + Mult[Iz, Sz], -1/2 ... 3 ..., ... 1 ...,
  1/2 i ω (g$I ((1 + 4 g$I^2 + 8 g$S^2) ω^2 - 2 ω ω$I + ω$I^2) Mult[I$m, aR] -
  g$I ((1 + 4 g$I^2 + 8 g$S^2) ω^2 - 2 ω ω$I + ω$I^2) Mult[I$p, aL] +
  g$S ω^2 Mult[S$m, aR] + 8 g$I^2 g$S ω^2 Mult[S$m, aR] + ... 70 ... +
  8 g$I^3 ω^2 Mult[I$m, Sz, aR, aR, aL] + 24 g$I g$S^2 ω^2 Mult[I$m, Sz, aR, aR, aL] -
  8 g$I (g$I^2 + 3 g$S^2) ω^2 Mult[I$p, Sz, aR, aL, aL] ) }, { ... 1 ... }, { ... 1 ... } }
```

large output

show less

show more

show all

set size limit...

Version 1

The cavity frequency is ω , atom I has resonance frequency ω (in tune with the cavity resonance), and atom S has resonance frequency $\omega - \Delta$ (detuned from the cavity resonance by an amount Δ). Here atoms I and S are different Species.

Take the coupling Hamiltonian into the interaction representation:

```
In[87]:= Clear[H$1$int];
```

```
H$1$int[t_] =
```

```
(Evolve[H$0, t, H$1] /. Evolve -> Evolver2) /. {ω$I -> ω, ω$S -> ω - Δ} //
```

```
Simplify // Expand
```

```
Out[88]:= g$I ω Mult[I$m, aR] + g$I ω Mult[I$p, aL] +
  e^{-i t Δ} g$S ω Mult[S$m, aR] + e^{i t Δ} g$S ω Mult[S$p, aL]
```

Using the Magnus expansion, compute a zeroth-order average Hamiltonian:

$$\text{In[89]:= } H_{\text{int}}^0 = \frac{1}{t_c} \text{Integrate}[H_{\text{int}}[t], \{t, 0, t_c\}] /. t_c \rightarrow \frac{2\pi}{\Delta} // \text{Expand}$$

$$\text{Out[89]= } g_I \omega \text{Mult}[I_m, a_R] + g_I \omega \text{Mult}[I_p, a_L]$$

To zeroth order, only atom *I* is coupled to the cavity.

Using the Magnus expansion, compute a first-order average Hamiltonian:

$$\text{In[90]:= } H_{\text{int}}^1 = -\frac{I}{2 t_c} \text{Integrate}[\text{Integrate}[\text{Comm}[H_{\text{int}}[t_1], H_{\text{int}}[t_2]], \{t_1, 0, t_2\}], \{t_2, 0, t_c\}] /. t_c \rightarrow \frac{2\pi}{\Delta}$$

$$\begin{aligned} \text{Out[90]= } & -\frac{1}{8\pi\Delta} i g_S \omega^2 \left(-4 i g_S \pi - 8 i g_S \pi S_z + 4 i g_I \pi \text{Mult}[S_m, I_p] + 4 i g_I \pi \text{Mult}[S_p, I_m] - \right. \\ & 16 i g_S \pi \text{Mult}[S_z, a_R, a_L] + 2 g_I \text{Mult}[I_m, S_p] \left(1 - \text{Cos}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] + \right. \\ & \left. \frac{2\pi \left(i \text{Cos}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \times \text{Sign}[\Delta] - \text{Sin}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right)}{\sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]} - \right. \\ & \left. i \text{Sign}[\Delta] \times \text{Sin}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right) + \frac{1}{\text{Abs}[\Delta]} \\ & 2 g_I \text{Mult}[I_p, S_m] \left(\frac{2\pi \left(i \text{Cos}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \times \text{Sign}[\Delta] + \text{Sin}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right)}{\sqrt{\frac{1}{\Delta^2}}} + \right. \\ & \left. \left. \text{Abs}[\Delta] \left(-1 + \text{Cos}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] - i \text{Sign}[\Delta] \times \text{Sin}\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right) \right) \right) \end{aligned}$$

Simplify the first-order average Hamiltonian by assuming that the resonance offset is positive.

Use sorted multiplication to put the *I* and *S* angular momentum operators in canonical order, with the *I*-atom operators first and the *S*-atom operators second.

```
In[91]:= ans1 = Expand[Simplify[H$1$int$1, Assumptions → {Element[Δ, Reals], Δ > 0}]] /.
{Mult → SortedMult}
```

$$\text{Out[91]} = -\frac{g^2 S^2 \omega^2}{2 \Delta} - \frac{g^2 S^2 S_z \omega^2}{\Delta} + \frac{g^2 I g^2 S \omega^2 \text{Mult}[I\$m, S\$p]}{\Delta} +$$

$$\frac{g^2 I g^2 S \omega^2 \text{Mult}[I\$p, S\$m]}{\Delta} - \frac{2 g^2 S^2 \omega^2 \text{Mult}[S_z, aR, aL]}{\Delta}$$

```
In[92]:= Collect[ans1, \frac{\omega^2}{\Delta}]
```

$$\text{Out[92]} = \frac{1}{\Delta} \omega^2 \left(-\frac{g^2 S^2}{2} - g^2 S^2 S_z + g^2 I g^2 S \text{Mult}[I\$m, S\$p] + \right.$$

$$\left. g^2 I g^2 S \text{Mult}[I\$p, S\$m] - 2 g^2 S^2 \text{Mult}[S_z, aR, aL] \right)$$

There are two physical effects contributed by the first-order terms.

1) A shift in the atom S energy levels, proportional to $(n + 1/2)$, with n the number of photons in the cavity.

2) A coupling between atoms I and S of the form $I_- S_+ + I_+ S_-$.

The coherence I_+ will evolve under this Hamiltonian into $I_+ \cos[\Omega t] + 2 i I_z S_+ \sin[\Omega t]$, entangling the electronic states of atoms I and S .

```
In[93]:= Evolve[Ω (Mult[I$m, S$p] + Mult[I$p, S$m]), t, I$p] /. Evolve → Evolver2
```

```
Out[93]= I$p Cos[t Ω] + 2 i Mult[Iz, S$p] × Sin[t Ω]
```

Version 2

The cavity frequency is ω and atom I and S each has a resonance frequency $\omega - \Delta$ (detuned from the cavity resonance by an amount Δ).

Here atoms I and S are the same species.

Take the coupling Hamiltonian into the interaction representation:

```
In[94]:= Clear[H$1$int];
```

```
H$1$int[t_] =
```

```
(Evolve[H$0, t, H$1] /. Evolve → Evolver2) /. {ω$I → ω - Δ, ω$S → ω - Δ} //
```

```
Simplify // Expand
```

$$\text{Out[95]} = e^{-i t \Delta} g^2 I \omega \text{Mult}[I\$m, aR] + e^{i t \Delta} g^2 I \omega \text{Mult}[I\$p, aL] +$$

$$e^{-i t \Delta} g^2 S \omega \text{Mult}[S\$m, aR] + e^{i t \Delta} g^2 S \omega \text{Mult}[S\$p, aL]$$

Using the Magnus expansion, compute a zeroth-order average Hamiltonian:

$$\text{In[96]:= H\$1\$int\$0} = \text{Expand}\left[\frac{1}{t\$c} \text{Integrate}[H\$1\$int[t], \{t, 0, t\$c\}] /. t\$c \rightarrow \frac{2\pi}{\Delta}\right]$$

Out[96]= 0

To zeroth order, the irradiation now has no effect.

The Hamiltonian does not commute with itself at different times.

$$\text{In[97]:= Comm}[H\$1\$int[t1], H\$1\$int[t2]]$$

$$\begin{aligned} \text{Out[97]=} & -e^{-i t1 \Delta + i t2 \Delta} g\$I g\$S \omega^2 \text{Mult}[I\$m, S\$p] + e^{i t1 \Delta - i t2 \Delta} g\$I g\$S \omega^2 \text{Mult}[I\$p, S\$m] - \\ & e^{-i t1 \Delta + i t2 \Delta} g\$I g\$S \omega^2 \text{Mult}[S\$m, I\$p] + e^{i t1 \Delta - i t2 \Delta} g\$I g\$S \omega^2 \text{Mult}[S\$p, I\$m] + \\ & e^{i t1 \Delta - i t2 \Delta} g\$I^2 \omega^2 \left(\frac{1}{2} + I_z + 2 \text{Mult}[I_z, aR, aL] \right) + \\ & e^{-i t1 \Delta + i t2 \Delta} g\$I^2 \omega^2 \left(-\frac{1}{2} + I_z - 2 (I_z + \text{Mult}[I_z, aR, aL]) \right) + \\ & e^{i t1 \Delta - i t2 \Delta} g\$S^2 \omega^2 \left(\frac{1}{2} + S_z + 2 \text{Mult}[S_z, aR, aL] \right) + \\ & e^{-i t1 \Delta + i t2 \Delta} g\$S^2 \omega^2 \left(-\frac{1}{2} + S_z - 2 (S_z + \text{Mult}[S_z, aR, aL]) \right) \end{aligned}$$

Using the Magnus expansion, compute a first-order average Hamiltonian:

$$\text{In[98]:= H\$1\$int\$1} = -\frac{I}{2 t\$c} \text{Integrate}[\text{Integrate}[\text{Comm}[H\$1\$int[t1], H\$1\$int[t2]], \{t1, 0, t2\}], \{t2, 0, t\$c\}] /. t\$c \rightarrow \frac{2\pi}{\Delta}$$

$$\begin{aligned} \text{Out[98]=} & -\frac{1}{8\pi\Delta} i \omega^2 \left(-4 i \pi (g\$I^2 (1 + 2 I_z) + g\$S^2 (1 + 2 S_z)) - \right. \\ & 2 g\$I (g\$S (\text{Mult}[I\$p, S\$m] + \text{Mult}[S\$p, I\$m]) + 2 g\$I \text{Mult}[I_z, aR, aL]) - \\ & 4 g\$S^2 \text{Mult}[S_z, aR, aL] + \\ & 2 (g\$I (g\$S (\text{Mult}[I\$m, S\$p] + \text{Mult}[S\$m, I\$p]) + 2 g\$I \text{Mult}[I_z, aR, aL]) + \\ & 2 g\$S^2 \text{Mult}[S_z, aR, aL]) - \\ & 2 i (g\$I (g\$S ((-i + 2\pi) \text{Mult}[I\$m, S\$p] + i (\text{Mult}[I\$p, S\$m] - \text{Mult}[S\$m, I\$p] + \text{Mult}[\\ & S\$p, I\$m]) + 2\pi (\text{Mult}[I\$p, S\$m] + \text{Mult}[S\$m, I\$p] + \text{Mult}[S\$p, I\$m])) + \\ & \left. 8 g\$I \pi \text{Mult}[I_z, aR, aL]) + 8 g\$S^2 \pi \text{Mult}[S_z, aR, aL]) \right) \end{aligned}$$

Use sorted multi\$plication to put the I and S angular momentum operators in canonical order, with the I-atom operators first and the S-atom operators second.

$$\text{In[99]:= ans2} = (H\$1\$int\$1 // \text{Expand}) /. \{\text{Mult} \rightarrow \text{SortedMult}\}$$

$$\begin{aligned} \text{Out[99]=} & -\frac{g\$I^2 \omega^2}{2\Delta} - \frac{g\$S^2 \omega^2}{2\Delta} - \frac{g\$I^2 I_z \omega^2}{\Delta} - \frac{g\$S^2 S_z \omega^2}{\Delta} - \frac{g\$I g\$S \omega^2 \text{Mult}[I\$m, S\$p]}{\Delta} - \\ & \frac{g\$I g\$S \omega^2 \text{Mult}[I\$p, S\$m]}{\Delta} - \frac{2 g\$I^2 \omega^2 \text{Mult}[I_z, aR, aL]}{\Delta} - \frac{2 g\$S^2 \omega^2 \text{Mult}[S_z, aR, aL]}{\Delta} \end{aligned}$$

$$\begin{aligned} \text{In[100]:= } & \text{Collect}\left[\text{ans2}, -\frac{\omega^2}{\Delta}\right] \\ \text{Out[100]= } & -\frac{\frac{g^2 I^2 \omega^2}{2 \Delta} - \frac{g^2 S^2 \omega^2}{2 \Delta} - \frac{\omega^2 (g^2 I^2 I_z + g^2 S^2 S_z + g^2 I g^2 S \text{Mult}[I\$m, S\$p] + g^2 I g^2 S \text{Mult}[I\$p, S\$m])}{\Delta}}{2 g^2 I^2 \omega^2 \text{Mult}[I_z, aR, aL]} - \frac{2 g^2 S^2 \omega^2 \text{Mult}[S_z, aR, aL]}{\Delta} \end{aligned}$$

There are three physical effects contributed by the first-order terms.

- 1) A shift in the atom S energy levels, proportional to $(n + 1/2)$, with n the number of photons in the cavity.
- 2) A shift in the atom I energy levels, proportional to $(n + 1/2)$, with n the number of photons in the cavity.
- 3) A coupling between atoms I and S of the form $I_- S_+ + I_+ S_-$.

Write an effective Hamiltonian assuming there are $n = 0$ photons in the cavity, so that

$$\begin{aligned} I_z a^\dagger a &\rightarrow I_z n \rightarrow 0 \\ S_z a^\dagger a &\rightarrow S_z n \rightarrow 0 \end{aligned}$$

Also set

$$\begin{aligned} g_I &\rightarrow g \\ g_S &\rightarrow g \end{aligned}$$

and write $I_{+/-}$ and $S_{+/-}$ in terms of the associated x and y angular momentum operators.

$$\begin{aligned} \text{In[101]:= } & \text{H\$eff} = \\ & \text{Simplify}\left[-\frac{\omega^2 (g^2 I^2 I_z + g^2 S^2 S_z + g^2 I g^2 S \text{Mult}[I\$m, S\$p] + g^2 I g^2 S \text{Mult}[I\$p, S\$m])}{\Delta}\right] /. \\ & \{ g^2 I \rightarrow g, g^2 S \rightarrow g, \\ & \quad I\$p \rightarrow I_x + I I_y, I\$m \rightarrow I_x - I I_y, \\ & \quad S\$p \rightarrow S_x + I S_y, S\$m \rightarrow S_x - I S_y \} \\ &]; \\ & \text{H\$eff} = \text{H\$eff} /. \frac{g^2 \omega^2}{\Delta} \rightarrow \Omega \end{aligned}$$

$$\text{Out[102]= } -\Omega (I_z + S_z + 2 \text{Mult}[I_x, S_x] + 2 \text{Mult}[I_y, S_y])$$

Let us calculate the evolution of the full density operator

$$\rho(0^+) = \left(\frac{1}{2} + I_x\right) \times \left(\frac{1}{2} + S_x\right)$$

This initial density operator can be created by applying a $\pi/2$ pulse to both atoms, cooled to their ground state.

$$\rho(0) = \left(\frac{1}{2} + I_z\right) \times \left(\frac{1}{2} + S_z\right)$$

$$\text{In[103]:= } \rho\$0 = \text{Mult}\left[\frac{1}{2} + I_z, \frac{1}{2} + S_z\right];$$

$$\text{In[104]:= } \rho\$init = \text{Evolve}\left[w (S_y + I_y), \frac{1}{w} \frac{\pi}{2}, \rho\$0\right] /. \text{Evolve} \rightarrow \text{Evolver2}$$

$$\text{Out[104]= } \frac{1}{4} + \frac{I_x}{2} + \frac{S_x}{2} + \text{Mult}[I_x, S_x]$$

Evolve this initial density operator under the effective Hamiltonian

$$\text{In[105]:= } \rho1 = (\text{Evolve}[H\$eff, t, \rho\$init] /. \text{Evolve} \rightarrow \text{Evolver1}) /. \text{Mult} \rightarrow \text{SortedMult} //$$

$$\text{Simplify} // \text{Expand}$$

$$\begin{aligned} \text{Out[105]= } & \frac{1}{4} + \frac{I_x}{4} + \frac{S_x}{4} + \frac{1}{4} I_x \cos[2 t \Omega] + \frac{1}{4} S_x \cos[2 t \Omega] + \cos[t \Omega]^2 \text{Mult}[I_x, S_x] + \\ & \text{Mult}[I_x, S_z] \sin[t \Omega]^2 + \text{Mult}[I_y, S_y] \sin[t \Omega]^2 + \text{Mult}[I_z, S_x] \sin[t \Omega]^2 - \\ & \frac{1}{4} I_y \sin[2 t \Omega] - \frac{1}{4} S_y \sin[2 t \Omega] - \frac{1}{2} \text{Mult}[I_x, S_y] \times \sin[2 t \Omega] - \\ & \frac{1}{2} \text{Mult}[I_y, S_x] \times \sin[2 t \Omega] + \frac{1}{2} \text{Mult}[I_y, S_z] \times \sin[2 t \Omega] + \frac{1}{2} \text{Mult}[I_z, S_y] \times \sin[2 t \Omega] \end{aligned}$$

Check that we recover the initial density operator at time zero.

We show below by looking at the $\rho1$ matrix, that the associated wavefunction is simply

$$|\uparrow\uparrow\rangle$$

which is what we expect for the system in its ground state.

$$\text{In[106]:= } \rho1 /. \{t \rightarrow 0\} // \text{Expand}$$

$$\text{Out[106]= } \frac{1}{4} + \frac{I_x}{2} + \frac{S_x}{2} + \text{Mult}[I_x, S_x]$$

Compute the post-interaction density operator at a time at which the equivalent pulse angle is $\pi/2$.

We will show below, numerically, that this is a maximally entangled state of the two qubits.

The post-interaction density operator corresponds to a wavefunction of

$$\frac{1}{2} \left(|\uparrow\uparrow\rangle - |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle - |\downarrow\downarrow\rangle \right)$$

which we determined with the help of Anthropic's Claude AI.

```
In[107]:=  $\rho1\$special = \rho1 /. \left\{t \rightarrow \frac{\pi}{2} \frac{1}{\Omega}\right\} // Simplify$ 
```

```
Out[107]=  $\frac{1}{4} + Mult[Ix, Sz] + Mult[Iy, Sy] + Mult[Iz, Sx]$ 
```

Load the matrices for two spin 1/2 particles.

```
In[108]:= << Matrices--two-spin-half.m
Loaded spin one-half matrices for mIx, mIy, mIz, mSx, mSy, mSz
```

Write the density operator in matrix form.

This is tricky. The “constant”, non-operator, part of the density operator should be replaced by the 4 x 4 unit matrix (i.e., IdentityMatrix[4]). In the code below we use the function ScalarPart[] to extract the scalar, *non-operator part* of the density operator. Multiply this scalar part by the symbol Id (which we will substitute with IdentityMatrix[4]) and add it back to the remainder of the density operator. Convert the result to matrix form using a set of substitutions.

```
In[109]:= rules1 = {Mult -> Dot};
rules2 =
  {Id -> IdentityMatrix[4], Ix -> mIx, Iy -> mIy, Iz -> mIz, Sx -> mSx, Sy -> mSy, Sz -> mSz};
```

```
ScalarPart[expr_] := Total[Select[List@@Expand[expr], FreeQ[#, _?OperatorQ] &]]
```

```
matrixify[rho_] := Module[{scalarp, rho$, rho$matrix},
  scalar = ScalarPart[rho];
  nonscalar = rho - scalar;
  rho$ = scalar * Id + nonscalar;
  rho$matrix = Simplify[(rho$ /. rules1) /. rules2];
  Print["Tr[ $\rho$ ] = ", Simplify[Tr[rho$matrix]]];
  Print["Tr[ $\rho^2$ ] = ", Simplify[Tr[rho$matrix . rho$matrix]]];
  Return[rho$matrix]
]
```

Verify by inspection that the matrixify[] function works correctly on the initial density operator.

The atomic subsystems should be in the ground-state wavefunction $|\uparrow\uparrow\rangle$ and we can verify that this is so by inspection.

We can also see that the initial density operator has unit trace, and its square has unit trace.

```
In[113]:=  $\rho\$0$ 
```

```
Out[113]=  $\frac{1}{4} + \frac{Iz}{2} + \frac{Sz}{2} + Mult[Iz, Sz]$ 
```

```
In[114]:=  $\rho\$0\$mat = matrixify[\rho\$0];$ 
 $\rho\$0\$mat // MatrixForm$ 
```

$$\text{Tr}[\rho] = 1$$

$$\text{Tr}[\rho^2] = 1$$

Out[115]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

As expected for a density operator, the special matrix also has unit trace and its square has unit trace.

In[116]:= `ρ1$special$mat = matrixify[ρ1$special];`

`ρ1$special$mat // MatrixForm`

$$\text{Tr}[\rho] = 1$$

$$\text{Tr}[\rho^2] = 1$$

Out[117]//MatrixForm=

$$\begin{pmatrix} \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ -\frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

A measure of the entanglement of two qubits is the value of the Wootters' concurrence function, doi:10.1103/PhysRevLett.80.2245.

The following concurrent-computing function was developed in collaboration with Anthropic's Claude AI.

```
In[118]:= Concurrence[rho_] := Module[{sy, rhoTilde, R, eigs},
  sy = {{0, -I}, {I, 0}};
  rhoTilde = KroneckerProduct[sy, sy].Conjugate[rho].KroneckerProduct[sy, sy];
  R = rho.rhoTilde;
  eigs = Sqrt[Sort[Re[Chop[Eigenvalues[R]]], Greater]];
  Max[0, eigs[[1]] - eigs[[2]] - eigs[[3]] - eigs[[4]]]
```

```
In[119]:= rho$test = {{0, 0, 0, 0}, {0, 1/2, -1/2, 0}, {0, -1/2, 1/2, 0}, {0, 0, 0, 0}};
ConcurrenceSymbolic[rho$test]
```

Out[120]= ConcurrenceSymbolic[$\left\{\left\{0, 0, 0, 0\right\}, \left\{0, \frac{1}{2}, -\frac{1}{2}, 0\right\}, \left\{0, -\frac{1}{2}, \frac{1}{2}, 0\right\}, \left\{0, 0, 0, 0\right\}\right\}$]

The initial density operator is a pure state, with a Wootters' concurrence of 0.

```
In[121]:= Concurrence[ρ$0$mat]
```

Out[121]= 0

The density operator following the timed interaction is a maximally entangled state, with a Wootters' concurrence of 1.

```
In[122]:= Concurrence[ρ1$special$mat]
```

```
Out[122]= 1
```

Let us calculate the Wootters' concurrence at any interaction time (i.e, pulse angle).

Rewrite the post-interaction density operator in terms of pulse angle instead of interaction time.

```
In[123]:= Clear[θ];
```

```
ρ1$mat = matrixify[ρ1] /. {t Ω → θ};
```

```
ρ1$mat // MatrixForm
```

```
Tr[ρ] = 1
```

```
Tr[ρ2] = 1
```

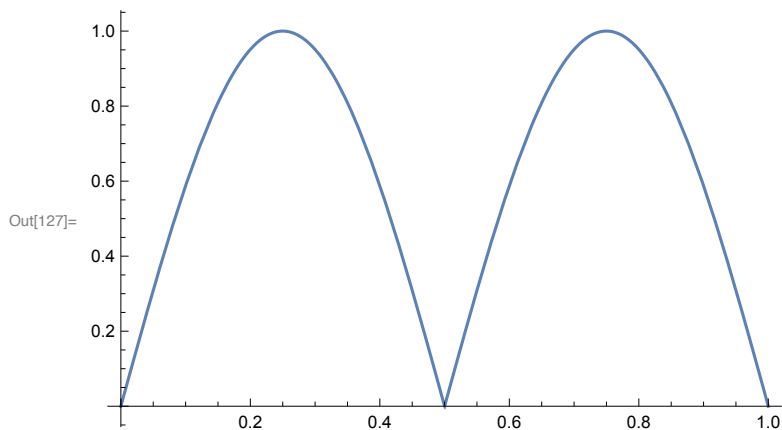
```
Out[125]//MatrixForm=
```

$$\begin{pmatrix} \frac{1}{4} & \frac{1}{4} (\cos[2\theta] - i \sin[2\theta]) & \frac{1}{4} (\cos[2\theta] - i \sin[2\theta]) & \frac{1}{4} (\cos[2\theta] - i \sin[2\theta]) \\ \frac{1}{4} (\cos[2\theta] + i \sin[2\theta]) & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} (\cos[2\theta] + i \sin[2\theta]) & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} (\cos[2\theta] + i \sin[2\theta]) & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

The function f computes the concurrence as function of pulse angle.

```
In[126]:= f[t_] := Concurrence[ρ1$mat /. {θ → 2 π t}]
```

```
In[127]:= Plot[f[t], {t, 0, 1}]
```



We can see that the concurrence is 1 when t is 0.25 and 0.75, corresponding to pulse angles of $\pi/2$ and $3\pi/2$.

```
In[128]:= {Concurrence[ρ1$mat /. {θ → π/2}],  
           Concurrence[ρ1$mat /. {θ → 3π/2}]}
```

```
Out[128]= {1, 1}
```

Clean up

```
In[129]:= resetVariables = False;  
(* Set to True to reset variable and to False to NOT reset the variables *)  
  
If[resetVariables,  
  Clear[Ix, Iy, Iz, I$P, I$m, Sx, Sy, Sz, S$p, S$m, aR, aL];  
  Clear[ $\omega$ ,  $\omega$ $I,  $\omega$ $S, g$I, g$S,  $\Delta$ , n];  
  Clear[H$0, H$1];  
]
```